

Frullani's Theorem

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0 Introduction

Theorem 0.1 (Frullani's Theorem). Suppose $f \in C[0, \infty)$, $\lim_{x \rightarrow \infty} f(x) = L$ and $a, b > 0$. Then:

$$\int_0^\infty \frac{f(bx) - f(ax)}{x} dx = (L - f(0)) \log\left(\frac{b}{a}\right)$$

Proof. If $a = b$, the result holds trivially as both sides are zero. Assume without loss of generality that $a < b$ (otherwise, switch a and b). Suppose $u, v > 0$ and $av > bu$. Then we have:

$$\begin{aligned} \int_u^v \frac{f(bx) - f(ax)}{x} dx &= \int_u^v \frac{f(bx)}{x} dx - \int_u^v \frac{f(ax)}{x} dx \\ &= \int_{bu}^{bv} \frac{f(x)}{x} dx - \int_{au}^{av} \frac{f(x)}{x} dx \\ &= \int_{av}^{bv} \frac{f(x)}{x} dx + \int_{bu}^{av} \frac{f(x)}{x} dx - \int_{bu}^{av} \frac{f(x)}{x} dx - \int_{au}^{bu} \frac{f(x)}{x} dx \\ &= \underbrace{\int_{av}^{bv} \frac{f(x)}{x} dx}_{G(v)} - \underbrace{\int_{au}^{bu} \frac{f(x)}{x} dx}_{G(u)} \end{aligned}$$

We now show that $\lim_{u \rightarrow 0^+} G(u) = f(0) \log\left(\frac{b}{a}\right)$. Suppose $\varepsilon > 0$. Then there exists $\delta > 0$ such that $|f(u) - f(0)| < \frac{\varepsilon}{\log(b/a)}$ for all $0 \leq u < \frac{\delta}{b}$ (and so $au, bu < \delta$). This yields:

$$\begin{aligned} \left| G(u) - f(0) \log\left(\frac{b}{a}\right) \right| &= \left| \int_{au}^{bu} \frac{f(x)}{x} dx - f(0) \log\left(\frac{b}{a}\right) \right| = \left| \int_{au}^{bu} \frac{f(x)}{x} dx - f(0) \int_{au}^{bu} \frac{1}{x} dx \right| \\ &= \left| \int_{au}^{bu} \frac{f(x) - f(0)}{x} dx \right| \leq \int_{au}^{bu} \frac{|f(x) - f(0)|}{x} dx \\ &\leq \frac{\varepsilon}{\log(b/a)} \int_{au}^{bu} \frac{1}{x} dx = \frac{\varepsilon}{\log(b/a)} \log\left(\frac{b}{a}\right) = \varepsilon \end{aligned}$$

Thus $\lim_{u \rightarrow 0^+} G(u) = f(0) \log\left(\frac{b}{a}\right)$.

We now show that $\lim_{v \rightarrow \infty} G(v) = L \log\left(\frac{b}{a}\right)$. Suppose $\varepsilon > 0$. Then there exists $H > 0$ such that $|f(x) - L| < \frac{\varepsilon}{\log(b/a)}$ for all $x > H$. This yields:

$$\begin{aligned} \left| G(v) - L \log\left(\frac{b}{a}\right) \right| &= \left| \int_{av}^{bv} \frac{f(x)}{x} dx - L \log\left(\frac{b}{a}\right) \right| = \left| \int_{av}^{bv} \frac{f(x)}{x} dx - L \int_{av}^{bv} \frac{1}{x} dx \right| \\ &= \left| \int_{av}^{bv} \frac{f(x) - L}{x} dx \right| \leq \int_{av}^{bv} \frac{|f(x) - L|}{x} dx \leq \frac{\varepsilon}{\log(b/a)} \int_{av}^{bv} \frac{1}{x} dx \leq \frac{\varepsilon}{\log(b/a)} \log\left(\frac{b}{a}\right) = \varepsilon \end{aligned}$$

Thus $\lim_{v \rightarrow \infty} G(v) = L \log\left(\frac{b}{a}\right)$. Finally, we have:

$$\int_0^\infty \frac{f(bx) - f(ax)}{x} dx = \lim_{v \rightarrow \infty} G(v) - \lim_{u \rightarrow 0^+} G(u) = (L - f(0)) \log\left(\frac{b}{a}\right)$$

A simpler proof, assuming f is differentiable and f' is integrable on $[0, \infty)$.

$$\int_0^\infty \frac{f(bx) - f(ax)}{x} dx = \int_0^\infty \int_a^b f'(xy) dx dy = \int_a^b \int_0^\infty f'(xy) dy dx = \int_a^b \frac{L - f(0)}{y} dy = (L - f(0)) \log\left(\frac{b}{a}\right)$$

Frullani's theorem was first stated¹ in 1821 in a letter from Giuliano Frullani (1795–1834) to Giovanni Plana (1781–

¹<http://www.ams.org/journals/proc/1990-109-01/S0002-9939-1990-1007485-4/S0002-9939-1990-1007485-4.pdf>

1864), though a complete proof was only given in 1823 by [Augustin-Louis Cauchy](#) (1789–1857).

Example. $f(x) = e^{-x}$

$$\int_0^\infty \frac{e^{-bx} - e^{-ax}}{x} dx = -\log\left(\frac{b}{a}\right)$$

Example. $f(x) = \arctan(x)$

$$\int_0^\infty \frac{\arctan(bx) - \arctan(ax)}{x} dx = \frac{\pi}{2} \log\left(\frac{b}{a}\right)$$

Example. $f(x) = \operatorname{arcsec}(x+1)$

$$\int_0^\infty \frac{\operatorname{arcsec}(bx+1) - \operatorname{arcsec}(ax+1)}{x} dx = \frac{\pi}{2} \log\left(\frac{b}{a}\right)$$

Example. $f(x) = \tanh(x)$

$$\int_0^\infty \frac{\tanh(bx) - \tanh(ax)}{x} dx = \log\left(\frac{b}{a}\right)$$

For certain values of a and b , this can be rewritten in other forms, e.g.

$$\tanh(2x) - \tanh(x) = \tanh(x) \operatorname{sech}(2x)$$

$$\tanh(3x) - \tanh(x) = \frac{2 \tanh(x) \operatorname{sech}(x)^2}{1 + 3 \tanh(x)^2}$$

$$\tanh(3x) - \tanh(2x) = \frac{\tanh(x) \operatorname{sech}(x)^4}{(1 + \tanh(x)^2)(1 + 3 \tanh(x)^2)}$$

Example. $f(x) = \operatorname{sech}(x)$

$$\int_0^\infty \frac{\operatorname{sech}(bx) - \operatorname{sech}(ax)}{x} dx = -\log\left(\frac{b}{a}\right)$$

For certain values of a and b , this can be rewritten in other forms, e.g.

$$\operatorname{sech}(x) - \operatorname{sech}(2x) = \frac{\operatorname{sech}(x)(1 - \operatorname{sech}(x))(2 + \operatorname{sech}(x))}{2 - \operatorname{sech}(x)^2}$$

$$\operatorname{sech}(x) - \operatorname{sech}(3x) = \frac{4 \operatorname{sech}(x) \tanh(x)^2}{4 - 3 \operatorname{sech}(x)^2}$$

Example. $f(x) = \arctan(\sinh(x))$

$$\int_0^\infty \frac{\arctan(\sinh(bx)) - \arctan(\sinh(ax))}{x} dx = \frac{\pi}{2} \log\left(\frac{b}{a}\right)$$

$$\begin{aligned} \arctan(\sinh(2x)) - \arctan(\sinh(x)) &= \arctan\left(\frac{\sinh(2x) - \sinh(x)}{1 + \sinh(x) \sinh(2x)}\right) \\ &= \arctan\left(\frac{2 \sinh(x) \cosh(x) - \sinh(x)}{1 + \sinh(x)(2 \sinh(x) \cosh(x))}\right) \\ &= \arctan\left(\frac{\sinh(x)(2 \cosh(x) - 1)}{1 + 2 \sinh(x)^2 \cosh(x)}\right) \end{aligned}$$

The last part of the proof of Frullani's theorem can be adapted to obtain a more general result:

Theorem 0.2. Suppose $f \in C[0, \infty)$, $\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x f(y) dy = L$ and $a, b > 0$. Then:

$$\int_0^\infty \frac{f(bx) - f(ax)}{x} dx = (L - f(0)) \log\left(\frac{b}{a}\right)$$

Example. $f(x) = \cos(x)$

$$\int_0^\infty \frac{\cos(bx) - \cos(ax)}{x} dx = -\log\left(\frac{b}{a}\right)$$

In this case, $\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x \cos(y) dy = \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} = 0$ while $\lim_{x \rightarrow \infty} \cos(x)$ does not exist.

Another way to evaluate this integral is to use Frullani's theorem with $f(x) = e^{-px} \cos(x)$ for $p > 0$, then take the limit as $p \rightarrow 0^+$. However, this requires care as the integral is not absolutely convergent for $p = 0$.

Example. Suppose $p, q \in \mathbb{R}$, $p > |q|$. Then we have:

$$\int_0^\infty \frac{\sin(px) \sin(qx)}{x} dx = \frac{1}{2} \int_0^\infty \frac{\cos((p-q)x) - \cos((p+q)x)}{x} dx = \frac{1}{2} \log\left(\frac{p+q}{p-q}\right) = \operatorname{arctanh}\left(\frac{q}{p}\right)$$

$$\int_0^\infty \frac{\operatorname{sech}(x)(1 - \operatorname{sech}(x))(2 + \operatorname{sech}(x))}{x(2 - \operatorname{sech}(x)^2)} dx = \log(2)$$

$$\int_0^\infty \frac{1}{x} \arctan\left(\frac{\sinh(x)(2 \cosh(x) - 1)}{1 + 2 \sinh(x)^2 \cosh(x)}\right) dx = \frac{\pi}{2} \log(2)$$

$$\int_0^\infty \frac{1}{x} \arccos\left(\frac{\cosh(4x) - \cosh(2x) + 2}{2 \cosh(x) \cosh(3x)}\right) dx = \frac{\pi}{2} \log(3)$$

$$\int_0^1 \frac{x}{(1 + 3x^2) \log\left(\frac{1+x}{1-x}\right)} dx = \frac{\log(3)}{4}$$

More examples are available at <https://scientia.mat.utfsi.cl/archivos/vol19/vol19art14.pdf>.

Theorem 0.3. Suppose $f \in C[0, \infty)$, $\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x f(y) dy = L$ and $a, b > 0$. Then:

$$\int_0^\infty \frac{f(bx) - f(ax)}{x} dx = (L - f(0)) \log\left(\frac{b}{a}\right)$$

Sub $u = e^x$ and $g(u) = f(\log(u))$, $\frac{1}{x} \int_1^{e^x} g(v)/v dv = L$

$$\int_1^\infty \frac{g(u^b) - g(u^a)}{u \log(u)} du = (L - g(1)) \log\left(\frac{b}{a}\right)$$

Sub $u = e^{-x}$ and $g(u) = f(-\log(u))$, $-\frac{1}{x} \int_1^{e^{-x}} f(v)/v dv = L$

$$\int_0^1 \frac{g(u^b) - g(u^a)}{u \log(u)} du = (L - g(1)) \log\left(\frac{b}{a}\right)$$

Corollary 0.4. Suppose $f \in C[1, \infty)$, $a, b > 0$ and $\lim_{x \rightarrow \infty} \frac{1}{\log(x)} \int_1^x \frac{f(y)}{y} dy = L$ exists and is finite. Then:

$$\int_1^\infty \frac{f(x^b) - f(x^a)}{x \log(x)} dx = (L - f(1)) \log\left(\frac{b}{a}\right)$$

Proof. Define $g : [0, \infty) \rightarrow \mathbb{R}$ by $g(u) = f(e^u)$. Then $g \in C[0, \infty)$. Substituting $u = \log(x)$ into the integral yields:

$$\int_1^\infty \frac{f(x^b) - f(x^a)}{x \log(x)} dx = \int_0^\infty \frac{f(e^{bu}) - f(e^{au})}{u} du = \int_0^\infty \frac{g(bu) - g(au)}{u} du$$

Similarly, substituting $u = \log(x)$ and $v = \log(y)$ into the limit condition yields:

$$\frac{1}{\log(x)} \int_1^x \frac{f(y)}{y} dy = \frac{1}{u} \int_1^{e^u} \frac{f(y)}{y} dy = \frac{1}{u} \int_0^u f(e^v) dv = \frac{1}{u} \int_0^u g(v) dv$$

Thus $\lim_{u \rightarrow \infty} \frac{1}{u} \int_0^u g(v) dv = \lim_{x \rightarrow \infty} \frac{1}{\log(x)} \int_1^x \frac{f(y)}{y} dy = L$. Applying **Theorem 0.2** to g yields:

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Suppose $f(x) = \sin(\log(x))$.

$$\int_1^x \sin$$

Suppose $\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x f(y) dy = L$. Substituting

$$\frac{1}{x} \int_0^x f(y) dy$$

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Proposition A.1.